

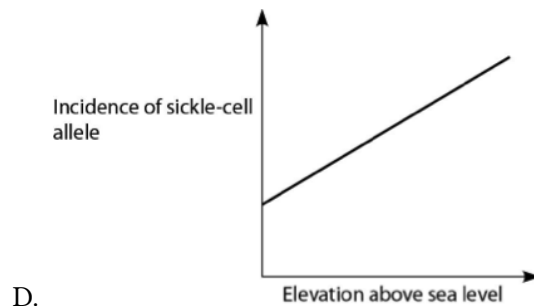
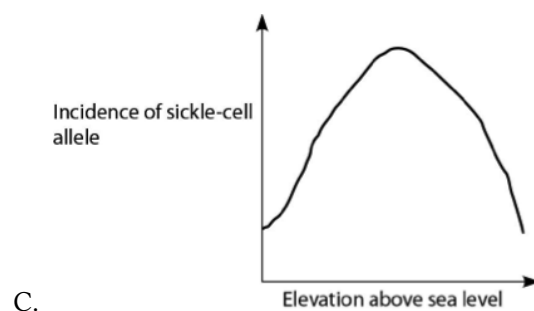
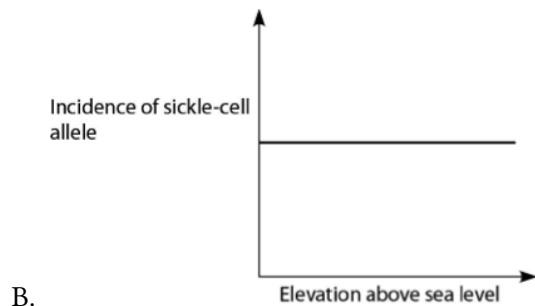
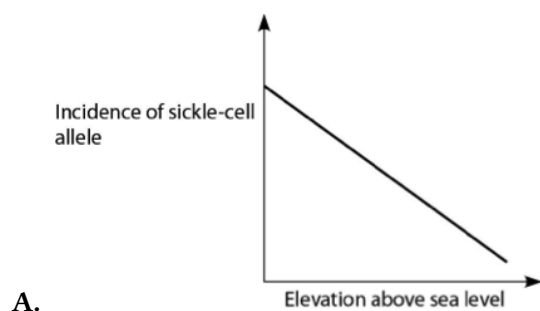
Chill. It's just a few pieces of paper. Each question is worth 2 points except where noted.

Circle True or False. Read each statement completely before answering.

1. True **FALSE** Thomas Robert Malthus wrote *Systema Naturæ* that influenced Darwin.
2. **TRUE** False The smallest unit of evolution is the population.
3. True **FALSE** One assumption of the Hardy-Weinberg equations is that mutations are random.
4. **TRUE** False Sexual selection usually causes a pattern of directional selection.
5. **TRUE** False Adaptations can lead to descent with modification.
6. True **FALSE** Overproduction of offspring by every individual in a population is necessary for evolution to occur.
7. **TRUE** False Genetic drift is strongest in small populations.
8. True **FALSE** The genotype is determined by the interaction of the phenotype with the environment.
9. **TRUE** False If $p = 0.4$, then total homozygosity in the population is 0.52 (assume equilibrium).
10. **TRUE** False Genetic drift always increases homozygosity in a population.

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

11. *Anopheles* mosquitoes, which carry the malaria parasite, cannot live above elevations of 5,900 feet. In addition, oxygen availability decreases with higher altitude. Consider a hypothetical human population that is adapted to life on the slopes of Mt. Kilimanjaro in Tanzania, a country in equatorial Africa. Mt. Kilimanjaro's base is about 2,600 feet above sea level and its peak is 19,341 feet above sea level. If the frequency of the sickle-cell allele in the population is plotted against altitude (feet above sea level), which of the following distributions is most likely, assuming little migration of people up or down the mountain?



12. Which *one* of the following statements is true of natural selection?
- A. It requires genetic variation.
 - B. It results in descent with modification and involves differences in reproductive success.
 - C. It involves differences in reproductive success.
 - D. It requires genetic variation, results in descent with modification, and involves differences in reproductive success.**
 - E. It results in descent with modification.
13. According to Darwin's concept of descent with modification:
- A. All organisms overproduce offspring, leading to differences in survivorship.
 - B. Individuals with greater fitness will evolve more quickly than individuals with lower fitness.
 - C. Species evolve from ancestral forms through the accumulation of different adaptations.**
 - D. Populations evolve by natural selection acting on individuals.
 - E. All organisms overproduce offspring, leading to different adaptations.
 - F. All species have evolved from a single common ancestor.
14. The overuse of drugs such as penicillin has caused an increase in the number of bacterial species that are resistant to the drugs. The most likely explanation for this result is:
- A. The drugs caused mutations in the bacteria that gave them resistance.
 - B. Bacteria began making proteins resistant to the drugs.
 - C. Penicillin was not an effective drug.
 - D. Most bacteria can resist drugs.
 - E. A few bacteria were naturally resistant and natural selection favored their increase.**
15. Which mouse has the greatest relative fitness? (A clutch is the offspring from one breeding event.)
- A. One that lives for 3 years, has 3 clutches per year, and 5 offspring per clutch.
 - B. One that lives for 3 years, has 3 clutches per year, and 7 offspring per clutch.**
 - C. One that lives for 2 years, has 3 clutches per year, and 8 offspring per clutch.
 - D. One that lives for 3 years, has 2 clutches per year, and 6 offspring per clutch.
 - E. One that lives for 4 years, has 3 clutches per year, and 4 offspring per clutch.
16. The rabbitsfoot mussel was once widespread throughout rivers of eastern North America but now remains in only a few very small populations. The decrease occurred rapidly due to habitat loss. Genetics studies show that the remaining populations have little or no genetic variation. The best explanation for this result is:
- A. Directional selection favored only one or two alleles.
 - B. Inbreeding caused an increase in heterozygosity.
 - C. Balancing selection favored only one or two alleles.
 - D. A bottleneck caused the loss of homozygosity.
 - E. Genetic drift caused the loss of heterozygosity.**
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17. Which of these conditions should completely prevent the occurrence of natural selection in a population over time?
- A. The population mates non-randomly.
 - B. The population size is large.
 - C. The population lives in a habitat where there are no competing species present.
 - D. The environment is changing at a relatively slow rate.
 - E. All variation between individuals is due only to environmental factors.**
18. Violation of these two Hardy-Weinberg assumptions always causes homozygosity to increase.
- A. Genetic drift and non-random mating.**
 - B. Natural selection and gene flow.
 - C. Genetic drift and natural selection.
 - D. Non-random mating and gene flow.
 - E. Mutation and natural selection.
 - F. Gene flow and mutation.
19. In science, a hypothesis is only useful if it
- A. can be tested.**
 - B. explains the natural world.
 - C. agrees with the conclusion.
 - D. is proven correct.
 - E. can be proven correct.
20. Organisms living in the northern rocky intertidal face competing selective forces. Larger individuals benefit by having a larger internal volume, which helps to conserve body temperature. However, smaller individuals are less likely to be dislodged by large waves from storms. Thus, selection may favor intermediate body size phenotype for any particular species. The mode of selection is
- A. natural selection.
 - B. stabilizing selection.**
 - C. phenotype selection.
 - D. disruptive selection.
 - E. directional selection.
21. (5 points) Here is the full classification of the Small-spotted Catshark.
- Eukaryota; Metazoa; Chordata; Chondrichthys; Carcharhiniformes; Scyliorhinidae; Scyliorhinus; canicula
- (a) Which is the class? **Chondrichthys**
- (b) Write the scientific name using proper binomial nomenclature. ***Scyliorhinus canicula***
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Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

adaptation	environment	Thomas Robert Malthus
allelotype	genotype	mutation
artificial selection	haploid	natural selection
balancing selection	independent evolution	phenotype
convergent evolution	inductive	population selection
Georges Cuvier	Jean Baptiste de Lamarck	scientific
deductive	Carolus Linnaeus	stabilizing selection
diploid	Charles Lyell	Alfred Russel Wallace

22. **Diploid** Has two alleles at a gene.
23. **Genotype** The combination of alleles at a gene.
24. **Lyell** Argued that Earth changed uniformly for much more than 6000 years.
25. **Convergent evolution** Unrelated organisms evolve similar traits by adapting to similar habitats.
26. **Adaptation** Trait that increases fitness in the current environment.
27. **Balancing Selection** Can cause stabilizing selection in a population.
28. **Lamarck** Was the first to propose that organisms change over time.
29. **Mutation** Creates new genetic variation within a population.
30. **Inductive** Reasoning that uses specific observations to develop general principles.
31. **Environment** Contributes to the phenotype but is not inherited.
32. Briefly compare sexual selection to natural selection to explain why sexual selection can cause evolutionary change more rapidly than other forms of natural selection.

Sexual selection acts directly on reproductive success while natural selection is indirect.

You may use a calculator to solve the following problem but you must **show your work** in the space provided. You may **round to 3 decimal places after each step or after the final calculation**. Any other rounding will result in a 3 point deduction.

33. (15 points) Impatiens is a type of flower that has three flower colors. Red color is produced by RR individuals. Pink color is produced by Rr individuals. White color is produced by rr individuals. A random sample of 251 flowers found that 107 were red, 33 were pink, and 111 were white. Calculate the two allele frequencies and the three genotype frequencies. *Show whether the population is in Hardy-Weinberg equilibrium.*

The final values may sum to 0.999, 1.000 or 1.001 due to rounding error. (2 pts each plus 5 points for showing your work.)

	Frequency
R	0.492
r	0.508
RR	0.426
Rr	0.131
rr	0.442

Be sure to show whether the population is in Hardy-Weinberg equilibrium and *show all work*.

Sums to 0.999.

34. (16 points) List below the four missing steps of the Scientific Method. Then, for each step, write a statement that illustrates how scientists applied the scientific method to the experiments described below. I have provided step four as an example.

Biologists wondered how homing pigeons are able return to their home roost from any direction and distance. One idea that had been proposed was that the pigeons orient using the Earth's magnetic fields, much like a compass. To test this idea, scientists built a flight chamber that was surrounded by a magnetic field. The scientists could adjust the field to change the apparent location of the magnetic north pole. After homing pigeons were placed in the chamber, the scientists changed the location of the apparent North Pole to several different locations around the chamber. In all cases, the pigeons changed their orientation to match the adjusted magnetic North Pole. Their results provided strong evidence that homing pigeons use the earth's magnetic fields for orientation.

- (a) **Observation:** Homing pigeons are able to return home from any direction and distance.
- (b) **Hypothesis:** Pigeons use the Earth's magnetic fields for orientation.
- (c) **Prediction:** If the orientation of the magnetic field is changed, then the birds will change accordingly.
- (d) **Experiment and results:** Scientists showed that homing pigeons use magnetic fields to orient their direction of flight.
- (e) **Conclusion:** The hypothesis was supported.

2 points for each step and 2 points for each explanation.
